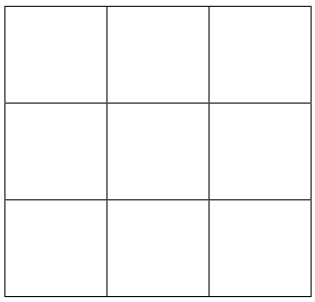
English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics / SEM Written examination, 17 August 2022 —— *Your grade will be 2 plus the number of points you receive. Time allowed: 3 hours. Good luck. Throughout, we work on a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$. *If needed, you may use* $1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$.

Problem 1. Suppose that every second a dart lands in a random square of the grid below.



- (0.5 points) What is the expected time until every square contains at least one dart?
- (0.3 points) What is the expected time until the first moment when some square contains two darts?
- (0.2 points) Can you generalize to an $n \times n$ grid?

Problem 2. Let $(X_1, \ldots, X_n)$ be a random permutation of the numbers in $\{1, 2, \ldots, n\}$, and let $S = X_1 + \cdots + X_n$.

1. (0.1 points) Find $\mathbb{E}S$ and DS.
2. (0.1 points) Prove that for two random variables X and Y, $D(X+Y) = DX + DY + 2Cov(X, Y)$.
3. (0.4 points) Express $\mathbb{E}S$ in terms of $\mathbb{E}X_i$. Find $\mathbb{E}X_i$, $\mathbb{E}X_i^2$ and DX_i for every i.
4. (0.4 points) Express DS in terms of DX_i and $Cov(X_i, X_j)$. Find $Cov(X_i, X_j)$ for all i, j.

Problem 3. Let $U_1, V_1, U_2, V_2, \ldots$ be independent $U(0,1)$ random variables, and let $M_i = \max(U_i, V_i)$.

- (0.75 points) Find $D(M_1 + \cdots + M_n)$ for every n.
- (0.25 points) Give an approximation to the distribution of $M_1 + \cdots + M_n$ for large n.

Problem 4.

1. (0.5 points) Let $X_1, \ldots, X_n$ be independent identically distributed random variables with density $f_X(x) = c/x^4$ for $x > 1$, and 0 otherwise. If $S_n = X_1 + \cdots + X_n$, find constants a, b, c such that $(S_n - a)/b$ is approximately standard normal, and justify your answer.
2. (0.5 points) Let $X_1, \ldots, X_{40}$ be independent $Poi(10)$ random variables. Find a constant a such that $X_1 + \cdots + X_{40} \in (a, \infty)$ with probability at least 99%.